

Last class, we learned that quantum operations must satisfy the following properties:

- **Unitary:** Given a superposition of n qubits

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle,$$

where α_x is the amplitude of the basic state $|x\rangle$ so that $\sum_{x \in \{0,1\}^n} \alpha_x^2 = 1$, a quantum operation U must be unitary, meaning that after applying U to $|\psi\rangle$, it is still true that $\sum_{x \in \{0,1\}^n} \alpha_x^2 = 1$.

- **Reversible:** A quantum operation U must be reversible, meaning that there must exist some quantum operation U^{-1} such that applying U and then U^{-1} to any superposition of n qubits $|\psi\rangle$ results in the original state $|\psi\rangle$.

For the following operations, either give their reversal operations or indicate that they are not reversible. We use capital letters to denote individual qubits; for example, if we have two qubits, we refer to the first qubit as A and the second qubit as B .

Operation U	Reversal Operation U^{-1}	Minimal example
$\neg A$	$\neg A$	$ 0\rangle \rightarrow 1\rangle \rightarrow 0\rangle$ $ 1\rangle \rightarrow 0\rangle \rightarrow 1\rangle$
if A then $\neg B$		
$A \oplus B$		
Hadamard A , defined as: $ 0\rangle \rightarrow \sqrt{\frac{1}{2}}(0\rangle + 1\rangle)$ (i.e, $\sqrt{\frac{1}{2}}$ amplitude on $ 0\rangle$ and $\sqrt{\frac{1}{2}}$ amplitude on $ 1\rangle$) $ 1\rangle \rightarrow \sqrt{\frac{1}{2}}(0\rangle - 1\rangle)$ (i.e, $\sqrt{\frac{1}{2}}$ amplitude on $ 0\rangle$ and $-\sqrt{\frac{1}{2}}$ amplitude on $ 1\rangle$)		